

Design of belt drives

(1) Velocity ratio.

D_1 = Diameter of driver

D_2 = Diameter of driving

N_1 = Speed of driver

N_2 = Speed of driven

length of the belt passing over driven in one minute

$$L = \pi D_1 N_1$$

length of the belt passing over driving in one minute

$$L = \pi D_2 N_2$$

no slip

$$\Rightarrow \pi D_1 N_1 = \pi D_2 N_2$$

$$\boxed{\frac{N_2}{N_1} = \frac{D_1}{D_2}}$$

when we consider thickness of belt.

$$\boxed{\frac{N_2}{N_1} = \frac{D_1 + t}{D_2 + t}}$$

$$V_1 = \frac{\pi D_1 N_1}{60} \text{ m/s}$$

$$V_2 = \frac{\pi D_2 N_2}{60} \text{ m/s}$$

for More drivers and driving.

$$\begin{aligned} \frac{N_2}{N_1} &= \frac{D_1 \times D_3}{D_2 \times D_4} = \frac{\text{Speed of last driver}}{\text{Speed of first driver}} \\ &= \frac{\text{Product of Dia of Drivers}}{\text{Product of Dia of followers}} \end{aligned}$$

(2) SLIP of the belt

Occurs when there is insufficient friction between belt and pulley causes forward motion of driver without carrying the belt with it is called slip.

Assuming let $s_1\%$ slip be present b/w driver & belt
 $s_2\%$ = follower & belt.

$$V = \frac{\pi D_1 N_1}{60} \rightarrow \frac{\pi D_1 N_1}{60} \times \frac{s_1}{100}$$

$$V = \frac{\pi D_1 N_1}{60} \left(1 - \frac{s_1}{100}\right)$$

Velocity of belt over follower

$$= \frac{\pi D_2 N_2}{60} \left(1 - \frac{s_2}{100}\right)$$

$$\frac{\pi D_2 N_2}{60} = V \left(1 - \frac{s_2}{100}\right)$$

$$\frac{\pi D_2 N_2}{60} = \frac{\pi D_1 N_1}{60} \left(1 - \frac{s_1}{100}\right) \left(1 - \frac{s_2}{100}\right)$$

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \left(1 - \frac{s_1 + s_2}{100}\right)$$

$$\boxed{\frac{N_2}{N_1} = \frac{D_1}{D_2} \left[1 - \frac{s}{100}\right]}$$

$$\boxed{\frac{N_2}{N_1} = \frac{D_{1+t}}{D_{2+t}} \left[1 - \frac{s}{100}\right]}$$

(3) Creep of belt

when the belt passes from slack side to the tight side a certain portion of the belt extends and contracts again when it passes from tight side to slack side.

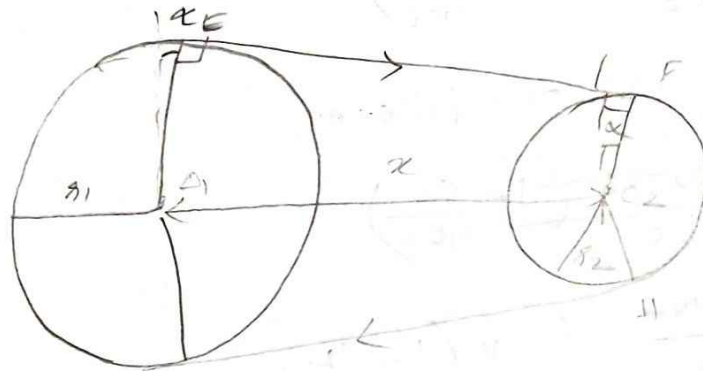
due to this changes of length, there is a relative motion between belt & pulley. This relative motion is called Creep.

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}$$

E = Young's Modulus.

σ_1, σ_2 = Stress on belt on tight & slack side

Length of open belt drive



$$\text{Length of the belt} = \text{Arc GJE} + EF + \text{Arc FKH} + HG$$

$$= 2 \times (\text{Arc JE} + EF + \text{Arc FK})$$

$$L = \pi(r_1 + r_2) + 2x + \frac{(r_1 - r_2)^2}{x}$$

Gross belt drive

$$L = \pi(r_1 + r_2)$$

Ratio of belt tension

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

Power transmitted $P = (T_1 - T_2) V$ watts

Centrifugal tension

$$T_c = m v^2$$

Maximum tension (T)

$$T = T_1 + T_c$$

Condition for Maximum power transmission

$$T_c = \frac{T}{3}$$

$$v = \sqrt{\frac{T}{3m}}$$

Q A flat belt is required to transmit 30 kW from a pulley of 1.5 m effective diameter running at 300 rpm. the angle of contact is spread over $\frac{11}{24}$ of circumference. the coefficient of friction between belt & pulley is 0.3. determine width of the belt, taking centrifugal tension, the given belt thickness is 9.5 mm.

Density = 1100 kg/m^3 & permissible working stress is 2.5 MN/m^2

$$P = 30 \text{ kW}, d = 1.5, N = 300 \text{ rpm}, \mu = 0.3$$

$$= 30000 \text{ W}$$

$$t = 9.5 \text{ mm}$$

$$P = (T_1 - T_2) v$$

$$30000 = (T_1 - T_2) \times \frac{\pi d N}{60}$$

$$T_2 (2.372 - 1) = 1273.23$$

$$30000 = (T_1 - T_2) \times \pi \times 1.5 \times \frac{300}{60}$$

$$T_1 - T_2 = 1273.23 \text{ N} \quad \text{--- (1)}$$

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

$$\theta = \frac{11}{24} \times 360^\circ = 165^\circ$$

$$= 2.88 \text{ rad}$$

$$\frac{T_1}{T_2} = e^{0.3 \times 2.88}$$

$$= T_1 = T_2 \times 2.372 \quad \text{--- (2)}$$

$$\underline{T_1 = 2200 \text{ N}}$$

$$\underline{T_2 = 927.5 \text{ N}}$$

width of belt, b

$$m = \text{Area} \times \text{length} \times \text{density}$$

$$m = w \times t \times l \times \rho$$

$$m = w \times 0.0095 \times l \times 1100$$

$$m = 10.45 b \text{ kg/m}$$

$$T_c = mv^2$$

$$T_c = m \left(\frac{\pi D N}{60} \right)^2$$

$$T_c = 10.45 b \times 23.57^2$$

$$T_c = 5805 b \text{ N} \quad \text{--- (2)}$$

$$T = T_1 + T_c = 2199 + 5805 b$$

$$T = \text{stress} \times \text{Area} = \sigma b t$$

$$2199 + 5805 b = 2.5 \times 10^6 \times b \times 0.0095$$

$$\underline{b = 0.122 \text{ m} = 122 \text{ mm}}$$

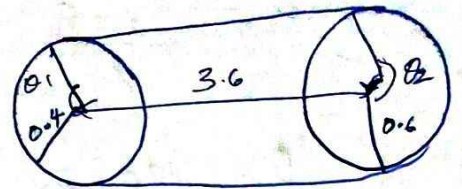
∴ Rounding to be standard = 125 mm.

- (8) Design a belt drive to transmit 110 kW for a system consisting of two pulleys of diameter 0.9 m and 1.2 m, center distance 3.6 m and a belt speed 20 m/s, coefficient of friction 0.3 slip 1.2% @ each pulley and 5% friction loss @ each shaft, 20% overload.
- Given, $P = 110 \text{ kW}$, $D = 0.9$, $D_2 = 1.2 \text{ m}$
 $\alpha = 3.6 \text{ m}$, $V = 20 \text{ m/s}$, $\mu = 0.3$, $S = 1.2\%$
 $F_f = 5\%$ at 20% overload

$$V = \frac{\pi D_1 N_1}{60} \left[1 - \frac{s_1}{100} \right]$$

$$20 = \frac{\pi \times 0.9 \times N_1}{60} \left[1 - \frac{1.2}{100} \right]$$

$$N_1 = ~~429~~ 430 \text{ rpm.}$$



$$V = \frac{\pi D_2 N_2}{60} \left[1 - \frac{s_1}{100} \right]$$

$$= \frac{\pi \times 1.2 \times N_2}{60} \left[1 - \frac{1.2}{100} \right]$$

$$N_2 = \underline{\underline{315.17}}$$

$$P = \frac{2\pi NT}{60}$$

$$T = \frac{P \times 60}{2\pi N_2} = \frac{\sqrt{10 \times 60}}{2 \times \pi \times 315} = ~~244.4~~ \underline{\underline{3334.6 \text{ Nm}}}$$

Considering friction loss Q .

$$T_1 = 1.05 \times T$$

$$= 1.05 \times 3334.6 = 3500 \text{ Nm.}$$

$$T_{\text{design}} = 1.2 \times T_1 = 4200 \text{ Nm}$$

Torque exerted on the driven pulley.

$$T = (T_1 - T_2) r_2$$

$$4200 = (T_1 - T_2) \times 0.6$$

$$(T_1 - T_2) = 7000 \text{ N, } \text{—————} \textcircled{1}$$

$$\sin \alpha = \frac{r_2 - r_1}{\alpha} = \frac{2.38}{\alpha} \text{ rad} = 2.4$$

$$\theta = 180 - 2\alpha = 180 - 4.8$$

$$= 175.2 = \underline{\underline{3.06 \text{ rad}}}$$

$$\frac{T_1}{T_2} = e^{\mu \theta}$$

$$T_1 = e^{0.3 \times 3.06} \times T_2 \quad \text{--- (2)}$$

$$\Rightarrow T_1 = 11636 \text{ N}$$

$$T_2 = 4636 \text{ N}$$

Assuming safe stress for the belt as 2.5 MPa . and thickness of belt is 5 mm . and belt speed is more than 10 m/s hence centrifugal tension is considered. Assuming density of belt as 1000 kg/m^3 .

$$m = A L \rho$$

$$= b t l \rho$$

$$\text{--- (1)}$$

$$b = \frac{1000}{156} \text{ --- (1)}$$

$$T_c = m v^2 = 15b \times 20^2$$

$$= 6000b \text{ N}$$

$$T_1 = T_2 + T_c = \sigma b t$$

$$= 116366 + 6000b = 2.5 \times 10^6 \times b \times 0.015$$

$$\underline{\underline{b = 400 \text{ mm}}}$$

Module 2

Design of spur gears

(1) Condition for constant velocity ratio of gears (law of gearing)

$$\frac{\omega_1}{\omega_2} = \frac{D_2}{D_1} = \frac{T_2}{T_1}$$

(2) Form of teeth.

1. Cycloidal
2. Involute

Standard proportion of gear system

Pg 229, table 12.3

Minimum no of teeth on the pinion to avoid interference.

$$T_p = \frac{2A_w}{G \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]}$$

G = Gear ratio / velocity ratio = $\frac{T_g}{T_p} = \frac{D_g}{D_p}$
 ϕ = pressure angle

Material of gear - 234 table 12-7

Design consideration for a gear drive

- 1) Power to be transmitted
- 2) speed of Driving gear.
- 3) speed of the driven gear
- 4) Center distance.

Lewis Equation: Beam strength of a gear

Eq 12.5b, 13.204

$$m = \sqrt{F_t / \sigma_d C_v K_y}$$

Permissible working stress for gear teeth in Lewis equation

$$\sigma_w = \sigma_a \times C_v$$

pg 205

Values of allowable static stress. p (234)
(from material table)

Dynamic tooth load. ^{tbl} pg 12.12 pg (236)

$$W_D = W_T + W_I$$

(Buckingham's eq)

$$= F_t + \frac{k_3 V (C_b + F_t)}{k_3 V + \sqrt{C_b + F_t}}$$

$$W = F_t$$

$$k_3 = 20$$

Dynamic factor

236, pg 12.12

$$W_f = \sigma_e b p_c Y = \sigma_e b \pi m Y = (\sigma_w) b \pi m Y$$

σ_e = flexural endurance limit

m = Module

p_c = Circular pitch

σ_y = strength factor.

b = width

Is taken as 3 pc to 4 pc for cut teeth
- 2 pc to 3 pc Cast teeth.

Wear tooth load.

$$W_w = D_p b Q K$$

D_p = Pitch circle diameter

b = face width.

Q = ratio factor



$$\phi = \frac{2 \times V_R}{V_R + 1}$$

$\Rightarrow$ external gear

$$\phi = \frac{2V_R}{V_R - 1}$$

$\rightarrow$ Internal gear.

databook 208 req (2.14 a).

load stress factor

$$K = \frac{\sigma_{es}^2 \sin \alpha}{1.4} \left[\frac{1}{E_1} + \frac{1}{E_2} \right]$$

pg (12.15(b)) ²⁰⁸

Causes of failure

- 1) Bending failure.
- 2) Pitting failure
- 3) scoring failure
- 4) Abrasion wear.
- 5) Corrosion wear.

Design procedure for Spur Gears

- 1) Design tooth load pg (204 - 12.4(b))
from power transmitted & pitch line velocity.

Relation : $W_T = \frac{P}{V} \times C_s$

W_T = Permissible tangential tooth load (N)

P = Power transmitted (W)

V = Pitch line Velocity in m/s $V = \frac{\pi D V}{60}$

D = Pitch circle Diameter (m)

N = Speed in rpm

C_s = Service factor — pg 235 table 12.8

Find static load by

$$W_s = \sigma_c b P_{cy} - \sigma_b \pi m y$$

$W_s > W_d$ (for safe design)

⑤ Find wear tooth load

$$W_w = D_p b Q K$$

W_w should not be less than W_d

Q

A Gear drive is required to transmit max power of 22.5 kW. The velocity ratio is 1:2 and the rpm of the pinion is 200 the approximate center distance between the shafts may be taken as 600 mm. The teeth has 20° stub involute profile, the static force for the gear material (cast iron) may be taken as 60 MPa and face width as 10 times the module. find the module, face width and number of teeth on each gear. check the design for dynamic and wear load. The deformation & dynamic factor is the buckingham & is taken as 80 and material combination factor for wear as 1.4:

$$P = 22.5 \text{ kW} = 22500 \text{ W} \quad \sigma_c = 60 \text{ MPa}$$

$$V = 1:2 \quad b = 10 \times M$$

$$n_{pmp} = 200 \quad C = 80$$

$$R = 600 \quad K = 1.4$$

$$\text{Angle} = 20^\circ$$

$$V_r = \frac{D_1}{D_2} = 2$$

$$V_r = \frac{D_1}{D_2} = \frac{1}{2}$$

$$D_2 = D_1 \cdot 2$$

$$600 = \frac{D_1}{2} + \frac{D_2}{2}$$

d_1 - pinion

d_2 = Gear

According to 12.3(a)

$$a = \frac{d_1 + d_2}{2} =$$

$$= 600 = \frac{d_1 + d_2}{2} = \frac{d_1 + 2d_1}{2}$$

$$\text{①} \quad d_1 = 400 \text{ mm}, d_2 = 800 \text{ mm}$$

Pitch line Velocity of pinion

$$V = \frac{\pi d_1 N_1}{60} = \frac{\pi \times 400 \times 200}{60} = \underline{\underline{4.2 \text{ m/s}}}$$

Velocity factor, C_v (805) 12.6(a) take only 3 instead of 3.5

$$C_v = \frac{3.05}{3.05 + V} = \frac{3.05}{3.05 + 4.2} = \underline{\underline{0.417}}$$

No. of teeth in pinion

$$T_p = \frac{D_p}{m} = \frac{400}{m}$$

Tooth form factor

$$Y_p = 0.175 - \frac{0.841}{T_p} = \underline{\underline{0.172}} \quad \text{--- } y_1$$

Equation for tangential tooth load

$$F_t = \frac{1000 P C_s}{v} \quad [12.7(a)] \quad , P \text{ is kW.}$$

$$= \frac{1000 \times 4.2 \times 1}{4.2} = \underline{\underline{5357}}$$

C_s is taken 1, for steady load, 8-10 hp/day

Lewis Equation

$$F_t = \pi \sigma_d C_v b y m$$

$$5357 = \pi \times 60 \times 0.417 \times 10 m \times (0.175 - 0.02 m) m$$

$$5357 = 1137.6 m^2 - 1.65 m^3$$

$$\underline{\underline{m = 8 \text{ mm}}}$$

$$b = 10 \times 8 = 80 \text{ mm}$$

$$T_p = \frac{400}{8} = 50$$

Module III - Design of Bearings or Journals

[349, 371]

Load

- (1) Radial Bearing
- (2) Thrust bearing.

Contact

- (1) Sliding Contact $\longrightarrow$ $\begin{cases} \text{full journal B (360)} \\ \text{Partial J-B} \\ \text{fitted J-B} \end{cases}$
- (2) Rolling Contact.

$\Rightarrow$ Thick film. (hydrodynamic B)
Thin
zero
Hydrostatic.

Assumptions hydrodynamic lubricated bearing.

- 1) Lubricant obeys Newton's law of Viscous flow.
- 2) Pressure is assumed to be constant throughout thickness.
- 3) Lubricant is assumed to be incompressible.
- 4) Viscosity is assumed to be constant.

Properties of sliding contact bearing materials

- Compressive strength
- $\rightarrow$ fatigue strength
- $\Rightarrow$ Conformability
- $\rightarrow$ Embeddability.
- $\Rightarrow$ ~~Reliability~~ bondability

Properties of lubricant

$\rightarrow$ Viscosity $= \frac{F}{a} = z \frac{dv}{dy}$

(15.1)

lubricating oil (363) (15.3) -

Dynamic load [Eq 12.12 pg 201]

~~Eq~~

$$F_d = \frac{F_t + k_3 v (C_b + F_t)}{k_3 v + \sqrt{C_b + F_t}}$$

$$= \frac{5357 + 20.67 \times 4.2 (80 + 5357)}{20.67 \times 4.2 + \sqrt{80 + 5357}}$$

$$= 10584.6 \approx \underline{\underline{10630 \text{ N}}}$$

Wear load (12.15a)

$$F_w = d_1 b Q K \quad , K = 1.4$$

$$Q = \frac{2d_2}{d_2 + d_1} = \frac{2 \times 800}{800 + 400} = 1.33$$

$$F_w = 400 \times 80 \times 1.33 \times 1.4 = \underline{\underline{58240 \text{ N}}}$$

$F_w > F_d \Rightarrow \text{Safe Design}$

$$F_T = \sigma_a b \pi m \gamma_p$$

$$\sigma_a = \text{table 12.5 (238)} = 83.5$$

$$= 83.5 \times 80 \times \pi \times 8 \times 0.175 - 0.021 \times 8$$

$$= \underline{\underline{25518.7}}$$

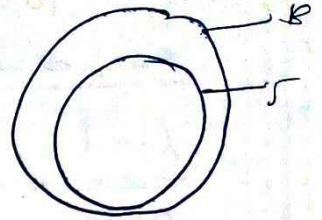
F_w is group

D = Dia of bearing

d = Dia of journal

L : length of bearing.

$$C = D - d$$



Diametral clearance ratio $\frac{D-d}{\phi}$

Eccentricity : radial distance b/w centre O of bearing and displaced centre O' of the bearing under load.

Short bearing $L/d < 1$

long bearing $L/d > 1$

square bearing $L/d = 1$

Sommerfeld no : $= \frac{ZN}{P}$

Design Value for journal bearing 369,
[15-11]

$$\text{Pressure of J.B} = P = \frac{1350}{10^6} ZN \left(\frac{d}{C} \right)^2 \left(\frac{1}{d+1} \right)$$

heat generated in a journal bearing.

heat dissipated : $Q_d = CA (t_b - t_a)$

Determine the bearing length by choosing ratio of L/d from table.
check bearing pressure $P = W/L.d$

Q6.1 Design a journal bearing for a centrifugal pump from the following data, load on journal = 20000 N
 Speed = 900 rpm. type of oil SAE 10, Absolute viscosity at 55°C = 0.017 kg/m/s Ambient temp of oil = 15.5°C
 Max pressure of bearing = 1.5 N/mm^2 . Calculate dia mass of lubricating oil required for artificial cooling if rise of temp is limited to 10°C . Heat dissipation coefficient = 123 W/m^2

eg 26.2

Ans length of bearing in mm.

$$A = l \times d = l \times 300.$$

Net allowable pressure.

$$1.6 = \frac{W}{A} = \frac{150 \times 10^3}{300 l} = \frac{500}{l}$$

$$\Rightarrow \underline{\underline{l = 312.5 \text{ mm}}}$$

Amount of heat removed.

$$q = \frac{32}{108} \text{ W}$$

eg 26.3 A full bearing of 50 mm dia and 100 long has a bearing pressure of 1.4 N/mm^2 ,
 $N = 900$, $\frac{\text{Journal dia}}{\text{dia clearance}} = 1000$, oil viscosity = 0.01 kg/m/s

(a) 75°C . room Temp = 35°C

- (1) find amount of cooling required
- (2) Mass of lubricating oil required if difference b/w outlet and inlet of oil is 10°C , C_p of oil = $1850 \text{ J/kg}^{\circ}\text{C}$

Ans: refer text

life of bearing.

Basic dynamic load rating of rolling bearing.

Dynamic equivalent load rating

$$W = X_V \cdot W_R + Y \cdot W_A$$

V = rotation factor

Dynamic load rating for rolling contact

$$L = \left(\frac{C}{W} \right)^k \times 10^6$$

$$C = W \left(\frac{L}{10^6} \right)^{1/k}$$

L = rating life.

C = Dynamic load bearing.

Q. 27.1 [Pg 1011]

A shaft rotating @ constant speed is subjected to variable load.

bearing support shaft are subjected to radial load of 3 kN for 10 percent of time, 2 kN for 20 percent of time, 1 kN for 30 percent of time. & no load for remaining time of cycle.

If total expected life is 20×10^6 revolutions at 95% reliability calculate dynamic load rating of ball bearing.